CHAPTER I

VARIATION UNDER DOMESTICATION

Causes of variability—Effects of habit and the use or disuse of parts —-Correlated variation—Inheritance—Character of domestic varieties—Difficulty of distinguishing between varieties and species—Origin of domestic varieties from one or more specks—Domestic pigeons, their differences and origin—Principles of selection, anciently followed, and their effects—Methodicat and unconscious selection—Unknown origin of our domestic productions—Circumstances favourable to man's power of selection.

Causes of Variability

WHEN WE COMPARE the individuals of the same variety or sub-variety of our older cultivated plants and animals, one of the first points which strikes us is, that they generally differ more from each other than do the individuals of any one species or variety in a state of nature. And if we reflect on the vast diversity of the plants and animals which have been cultivated, and which have varied during all ages under the most different climates and treatment, we are driven to conclude that this great variability is due to our domestic productions having been raised under conditions of life not so uniform as, and somewhat different from, those to which the parent species had been exposed under nature. There is, also, some probability in the view propounded by Andrew Knight, that this variability may be partly connected with excess of food. It seems clear that organic beings must be exposed during several generations to new conditions to cause any great amount of variation; and that, when the organisation has once begun to vary, it generally continues varying for many generations. No case is on record of a variable organism ceasing to vary under cultivation. Our oldest cultivated plants, such as wheat, still yield new varieties: our oldest domesticated animals are still capable of rapid improvement or modification.

As far as I am able to judge, after long attending to the subject, the conditions of life appear to act in two ways,—directly on the whole